Jahid Md Hasan

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Summary

AI/ML engineer with an MCA in Computer Science and hands-on experience building LLM-powered applications, retrieval-augmented generation systems, AI agents and machine learning pipelines. Experienced in Python development, retrieval diagnostics, automated testing, model experimentation and debugging. Interested in building reliable, measurable AI systems and improving their performance through systematic evaluation.

Work Experience

Machine Learning Intern – Unified Mentor June 2025 – Sep 2025

- Built end-to-end ML pipelines across healthcare, finance, and accessibility domains, covering data preprocessing, feature engineering, model training, and deployment workflows.
- Developed a fraud detection system on temporal transaction data (XGBoost, Random Forest, imbalanced classification) and a heart-disease diagnostic model with SHAP/LIME interpretability, evaluated via ROC-AUC and precision-recall.

Data Science Intern – InLighnX Global Oct 2025 – Feb 2026

- Built an air-quality forecasting system using Random Forest, Prophet, and LSTM, with automated preprocessing, feature engineering, model evaluation, and an interactive dashboard.
- Developed an AI-powered voice interaction system integrating speech recognition, NLP-based command routing, and event-driven automation workflows.

Projects

DocuMind AI – Agentic RAG Document Assistant View in GitHub

- Engineered a document assistant with an LLM agent loop that selects PDF retrieval, calculator and summarisation tools; supports OpenAI/Groq provider switching through LangChain.
- Built a PDF-based RAG pipeline using PyMuPDF, 500-token overlapping chunks, all-MiniLM-L6-v2 embeddings, and persistent ChromaDB, achieving sub-200ms CPU retrieval; deployed a multi-turn Gradio chat interface with conversational memory and source citations.
- Established a reproducible 15-query retrieval baseline: Hit@1 66.7%, Hit@3 86.7%, Hit@5 93.3%, MRR 0.783 and 24.4 ms average retrieval latency.

Obsidian Vault RAG Assistant View in GitHub

- Built a Markdown knowledge-base assistant with heading-aware chunking, MiniLM embeddings, dense cosine retrieval and custom BM25 lexical search.
- Fused rankings with Reciprocal Rank Fusion; added a relevance gate, grounded generation, source citations and a deterministic evaluation suite.

Network Security Log Triage Agent View in GitHub

- Implemented modular security-ML workflows with MongoDB ingestion, schema validation, KS-test drift detection, model training, MLflow tracking, FastAPI inference and Docker.
- Investigated dataset reproducibility, duplicate/leakage risks and evaluation protocol design; agentic investigation, RAG and MCP capabilities remain planned extensions.

Real-Time Laser Boundary Detection View in GitHub

- Built a modular OpenCV pipeline (segmentation, morphology, contour extraction, clustering, geometric fitting, temporal tracking) to detect and track forklift laser safety perimeters.
- Designed deterministic CPU-bound inference for edge deployment, using geometric constraints and tracking to suppress detection jitter across real-time footage.

Education

Amity University – Noida Sep 2023 – March 2026
Master of Computer Applications (MCA) – Specialization in AR/VR & Intelligent Systems

Assam University – Silchar Aug 2019 – Nov 2022
Bachelor of Science (BSc.) – Computer Science Honours

Skills

Languages & APIs: Python, SQL, C++, JavaScript, FastAPI, REST APIs, Git, Linux
AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, Transformers
GenAI & Agents: LangChain, LangGraph, CrewAI, RAG, Agentic & Multi-Agent Workflows, MCP, Tool Calling, Prompt Engineering, Fine-Tuning
Retrieval & Eval: FAISS, Chroma, LlamaIndex, RAGAS, LangSmith, Embeddings, Semantic Search, BM25, RRF, relevance gating, retrieval metrics, source citations
Computer Vision: OpenCV, Image Processing, Real-Time Video Pipelines, Object Detection Fundamentals
MLOps & Cloud: Docker, AWS, MLflow, CI/CD, Drift Detection, ETL Pipelines, MongoDB Atlas
Certifications: IBM Data Science | TCS AI Pro | IBM RAG & Agentic AI | RoboAI Industrial Training